

You Are Here: When the Place a Task Is Stuck Predicts What the Problem Feels Like

Human-readable companion to Bottleneck-Relative Organization of Task-Relevant Phenomenal Content: The You Are Here Hypothesis

Human-readable companion to the completed scientific manuscript. This version preserves the scientific claims and limits of the completed manuscript while presenting the motivating problem and argument in more direct language. It does not constitute a separate research contribution.

Abstract

A goal-directed task can contain many active processes at once: generating possibilities, choosing among them, turning a choice into an executable plan, monitoring what happens, and revising when necessary. Yet conscious task experience is often much narrower than that full architecture. The You Are Here hypothesis asks whether the part of the action system that is currently limiting progress predicts what the focal experienced problem is about. If generation is limiting, experience may be organized around missing possibilities; if selection is limiting, around unresolved alternatives; if implementation is limiting, around how to carry out an already chosen option. The proposal is not that consciousness literally moves through layers or that impasse creates consciousness. It is a cross-level predictive correspondence between an independently identified functional bottleneck and task-relevant focal phenomenal content. The hypothesis remains empirically undemonstrated. A valid test must distinguish the predicted content shift from generic difficulty, attention, metacognitive feelings, explicit task-state diagnosis, demand characteristics, and report reconstruction, and it must allow the hypothesis to shrink or fail if those alternatives explain the result.

1. The human problem: the whole task exists, but one part becomes “the problem”

Think about a difficult task you already understand. The goal may be clear, the environment may be unchanged, and the same overall action system may remain active. What changes from moment to moment is what is holding progress up. At one point you may have no viable idea what to do. Later you may have several plausible options but cannot choose. Later still you may know exactly what you want to do but not how to execute it.

Those states can all be called “stuck,” yet they are not functionally identical. More importantly, the focal conscious problem also seems capable of changing its subject: no ideas, too many competing alternatives, missing implementation detail, uncertain feedback. The question is whether that apparent correspondence is systematic enough to be a scientific object rather than an intuitive description.

You Are Here takes its name from that possibility. The proposed “location” is functional location within a goal-directed architecture, not a place in the brain and not a literal layer through which consciousness travels. The manuscript asks whether where progression is currently limited predicts what task-relevant focal experience is about.

2. The action system has more structure than focal experience

Established work already distinguishes important parts of goal-directed action. Options can be generated before they are selected; candidate actions can compete; selected goals can require further specification; monitoring and feedback can reopen earlier work; and control can be hierarchical or recurrent rather than a clean one-way pipeline (Cisek, 2007; Kaiser et al., 2013; Kalis et al., 2013; Botvinick, 2008).

Problem-solving research likewise shows that impasse is not one homogeneous state. Cognitive architectures such as Soar localize impasse relative to the operation that cannot proceed, while human problem-solving research distinguishes stages and restructuring processes (Laird, 2012; Knoblich et al., 1999; Fedor et al., 2015). None of this, by itself, demonstrates the You Are Here correspondence.

The starting observation is therefore not that the action architecture is new. It is that conscious task experience is selective relative to it. The larger system may still contain the goal, candidate set, choice relation, implementation

requirements, predictions, and feedback simultaneously while focal experience gives disproportionate prominence to only part of that structure.

3. The proposed correspondence

Let B denote the functionally dominant bottleneck: the relation currently limiting progression. Let ΦF denote task-relevant focal phenomenal content. Let X and A collect relevant contextual and architectural controls. The minimal proposal is that bottleneck identity contains predictive information about focal phenomenal organization beyond those controls:

$$I(B ; \Phi F \mid X, A) > 0$$

The stronger dynamic prediction asks what happens when the bottleneck itself moves while a substantially common task architecture is preserved:

$$B_i \rightarrow B_j \Rightarrow \Phi_i \rightarrow \Phi_j$$

That shorthand is distributional, not deterministic. It does not predict that every single report flips cleanly whenever one internal relation changes. It predicts that the distribution or referential organization of task-relevant focal content should shift when the dominant limiting relation shifts.

The proposal concerns identity more than intensity. Merely becoming more difficult, effortful, uncertain, slow, or conflict-laden is not enough. A generation-limited state and a selection-limited state might be equally difficult while differing in what the unresolved problem is about.

4. What the prediction might look like in ordinary language

Dominant limiting relation	Possible focal problem	What must not be inferred automatically
Generation	"I cannot find a viable possibility."	That option generation is the only active process or a unique neural mechanism
Selection	"I have alternatives, but I cannot settle which one."	That conscious indecision transparently reveals the causal architecture
Specification / implementation	"I know which option I want; I cannot work out how to do it."	That implementation is literally a later conscious layer
Monitoring / feedback	"Something about what is happening is not matching what I need."	That all monitoring becomes conscious or that feedback difficulty has one mechanism

These examples explain the hypothesis; they do not define successful evidence. The content categories themselves are measurement constructs. In a real experiment, report structure should be derived and coded independently of condition labels. If the data organize differently from the illustrative categories, the data should win.

5. Why "people report different things" would not be enough

The hardest inferential problem is that reports can be generated after the probe. A participant may know, from the visible task state, that there were several options and then answer "I was deciding" even if selection-related content was not what had been phenomenally focal immediately beforehand. A task can therefore create the predicted language without creating the predicted online experience.

For that reason the manuscript distinguishes the theoretical phenomenal target ΦF from coded immediate report content RF . Reports are evidence relevant to experience, not direct access to experience. They can omit content, compress it into language, perturb the state being measured, or recruit post-prompt reconstruction.

The empirical strategy therefore treats semantic reconstruction as a competitor rather than a nuisance to be waved away. Neutral, sparse, immediate probes should be combined with explicit task-state diagnosis and semantic-yoked controls. A strong result requires the bottleneck-relative referential pattern to exceed what participants can generate simply from knowing the local task state.

$$\text{observable report pattern } RF \neq \text{phenomenal state } \Phi F$$

If explicit diagnosis or semantic-yoked controls reproduce the same differentiation, the experiment may show bottleneck-relative reportable problem representation, but it does not identify the stronger phenomenal correspondence.

6. The alternatives the hypothesis has to beat

Several mature explanations can generate nearby observations. Difficulty, effort, uncertainty, conflict, response latency, surprise, familiarity, evidence strength, and attentional capture can all change with bottleneck manipulations. Global-access or cognitive-control accounts can explain why unresolved nonroutine processing becomes salient or widely available (Baars, 2002; Dehaene & Naccache, 2001; Botvinick et al., 2001).

Metacognitive-feeling accounts are especially close. People can experience difficulty without correctly diagnosing its source, and subjective difficulty can depend on multiple cues (Desender et al., 2017, 2021; Fernández Velasco & Loev, 2025). You Are Here retains residual value only if independently validated bottleneck identity or bottleneck relocation adds structure beyond those process-cue accounts.

Phenomenological field theories and affordance-based approaches also precede the idea that conscious experience is structured rather than a flat list of contents (Gurwitsch, 1964; Yoshimi & Vinson, 2015; Rietveld et al., 2018; Proust, 2024). The manuscript does not claim to discover structured phenomenal fields. Its OPEN contribution is the specific proposed correspondence between a functionally limiting relation and the organization of task-relevant focal content.

7. A test has to move the bottleneck without simply changing the whole task

A useful experiment keeps an overarching task architecture substantially common while making different relations dominant limits. The RGP_SubV2 manuscript gives a worked route-planning example: the same broad goal, interface, response method, route workspace, and execution controls can be retained while different trials are designed to make candidate generation, selection, or implementation the dominant unresolved relation.

Bottleneck identity has to be validated before analyzing the phenomenal report. A generation-limited condition might require delayed first viable candidate and show selective rescue when a valid candidate is supplied. A selection-limited condition should contain at least two independently viable candidates, prolonged or revisited competition, and selective release when one viable candidate is externally designated. A specification-limited condition should show a stable selected route with elevated choice-to-executable-plan latency or implementation errors, and selective release when missing mapping information is provided without changing the selected route.

If two candidate bottlenecks cannot be distinguished by task structure and preregistered behavioral criteria before report analysis, the theory does not get to keep the fine labels. They must be collapsed to the coarsest independently identifiable distinction or the ambiguous trials must be excluded.

8. The dynamic test is the strongest version

Cross-sectional differences are useful but vulnerable to semantic and demand confounds. The strongest test moves the dominant bottleneck within a substantially common architecture and asks whether the distribution of focal content relocates with it.

A semantic-yoked comparison is especially important. Two conditions can preserve similar visible route cards and choice-oriented language while changing which relation is actually limiting. For example, selection can be made easy while implementation mapping is withheld. If immediate report content follows the validated implementation bottleneck rather than the surface choice semantics, the semantic-reconstruction account loses some explanatory ground.

The empirical burden remains asymmetric. A clean relocation that survives the controls would support the correspondence. Failure of focal content to relocate when the bottleneck moves is direct evidence against the strongest dynamic form.

9. Clear outcomes that shrink or reject the proposal

Outcome	Interpretation
Only generic difficulty/effort changes	Bottleneck identity adds no referential organization beyond demand.
Reports differ, but semantic/task-state controls reproduce the differences	Supports reportable task-state reconstruction, not the stronger phenomenal claim.
Only delayed explanations differentiate conditions	Retrospective reconstruction is favored over online focal organization.
Bottleneck identity cannot be independently validated	Architectural test fails; ΦF cannot rescue an unidentified B.
Metacognitive-process accounts fully absorb the report structure	Use the mature account; residual You Are Here value contracts.
Bottleneck moves but focal-content distribution does not	Strong dynamic You Are Here prediction fails.
Candidate bottleneck distinctions cannot be separated	Collapse them to the coarser identifiable factor.

10. What the hypothesis does not say

You Are Here does not require people to understand the architecture generating their experience. A person can experience uncertainty about alternatives without correctly diagnosing “selection” as the limiting relation. Explicit metacognitive diagnosis and experienced content are different measurement objects.

The functional relations need not form a serial chain. Generation, selection, specification, monitoring, and revision can be recurrent, hierarchical, or partially parallel. “Location” means functional location within the organized action system, not a compulsory temporal stage.

The hypothesis does not assert one bottleneck at every moment. Multiple relations can simultaneously constrain progress; the initial test deliberately targets conditions where one dominant bottleneck is identifiable. Ambiguous multi-bottleneck episodes cannot be used to rescue a failed clean test.

The model is not a neural theory of consciousness, does not claim that a bottleneck causes consciousness, does not determine the relation between attention and phenomenality, and does not explain why consciousness exists or why it has qualitative character. It proposes a predictive cross-level correspondence under specified conditions.

11. Why this matters for subjectivity without turning it into a theory of agency

If the correspondence is supported, the larger action system could contain a richer causal map than focal experience exposes. Goals, candidate generation, competition, implementation, prediction, and feedback could all contribute to behavior while conscious task content preferentially reflects the locally unresolved relation.

That would make some conscious uncertainty local rather than global. A person may know the goal but not which action to select, or know the intended action but not how to implement it. Focal experience could then reveal something about where progression is unresolved without revealing the full causal history of action.

That implication remains conditional. The manuscript does not establish that agency is organized this way, and possible applications to ADHD or other individual differences require separate evidence. Those domains are not evidence for the general hypothesis merely because their phenomenology seems compatible.

12. Conclusion

The You Are Here hypothesis asks one constrained question: when the functional location of unresolved progression changes inside a substantially common goal-directed architecture, does what the focal conscious problem is about change with it?

A convincing answer requires more than differentiated self-report. The bottleneck must be identified independently. Report content must be sampled close to the unresolved episode. Generic demand, metacognitive feelings, explicit task-state diagnosis, and semantic reconstruction must be allowed to explain the pattern first. Fine functional distinctions must collapse when they cannot be validated.

The proposal therefore succeeds only if bottleneck identity adds predictive organization to task-relevant focal content beyond those alternatives, with the strongest evidence coming from a content relocation when the bottleneck itself moves. If the content stays put, or if mature alternatives reproduce the pattern, the strong correspondence should be reduced or rejected.

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